Xingye Tan

U.S. WORK AUTHORIZATION (NO SPONSORSHIP NEEDED)

719-321-4659 xtan4@uw.edu linkedin.com/in/xingye-tan-817b7a225 github.com/AidenTan-DS

Education

University of Washington **Seattle, WA**

MS in Data Science Expected March 2027

- **Scholarship:** $20,041 merit-based Merit & Opportunity Scholarship (2025–26)
- **Coursework:** Statistical Machine Learning, Applied Statistics & Experimental Design, Scalable Data Systems and Algorithms, Software Design for Data Science

Pennsylvania State University **University Park, PA**

BS in Mathematics (Computational) and Statistics (Statistics & Computing) | GPA: 3.85/4.0 Aug 2021 – May 2025

Skills

Languages & Frameworks: Python (PyTorch, JAX, TensorFlow, scikit-learn, Pandas, NumPy), SQL (MySQL), R
ML & AI: speech recognition (RNN-T, Conformer), CNN, RNN, ensembles (XGBoost, Random Forest), LLM APIs, RAG, MLflow, IBM Bob (agentic AI coding assistant)
ML Systems & Data: distributed multi-GPU training, LSF cluster scheduling, performance profiling, PySpark, Databricks, AWS (S3, EC2), Docker, Git/GitHub, CI/CD (GitHub Actions)

Work Experience

AI/ML Engineer Intern **Jun 2026 – Sep 2026**

IBM | watsonx Assistant, Speech Team San Jose, CA

- **Model Training & Evaluation:** Filtered **5.69M** corpus entries (SNR, VAD, transcript checks) into **4.15M** training samples (**5,030+ hours**), trained a **Streaming Conformer RNN-T** on distributed GPUs, and evaluated it across **15 benchmarks**, reducing average WER from **66.9% to 22.4%** versus the raw baseline
- **Reliability:** Found a silent failure where training ran with normal loss but only **4 of 336** pretrained tensors loaded; fixed it to **336 of 336** and added end-to-end smoke tests, checkpoint parameter verification, and CPU liveness monitoring
- **Performance Optimization:** Cut a feature-extraction workload from an estimated **5.3 hours to 78 minutes** by tracing CPU contention to a mismatch between LSF-requested cores and framework threads, then unifying core allocation with BLAS thread configuration
- **Experimentation:** Ran **4 fine-tuning experiments** with synthetic TTS data; evaluation showed degraded generalization to real conversational speech, so the approach was stopped
- **ML Infrastructure:** Resolved recurring issues on IBM's compute cluster, including **LSF scheduling**, **multi-GPU distributed training**, filesystem I/O, node failures, Python/Conda environments, and data corruption

Research Experience

Research Support **Jun 2024 – May 2025**

Penn State, Department of Statistics | Python, LLMs *GitHub*

- **Data Pipeline & Monitoring:** Built an **ETL pipeline** with LLM API integration to parse 22K+ unstructured clinical trial records at a **93% parsing rate**, with a **5x query speed** improvement and **drift monitoring** for data quality
- **Matching System:** Built a hybrid matching system (fuzzy matching, edit distance, Word2Vec embeddings) that raised treatment-protocol matching accuracy from **63% to 95%** and cut manual review by **85%**
- **Recommendation Knowledge Base:** Delivered a **RAG-ready knowledge base** with cross-functional validation for automated treatment recommendations; awarded **3rd Place, Penn State Data Science Poster Competition**

Projects

Lumbar Spine Degenerative Classification | Python, PyTorch, MLflow **May 2024 – Oct 2024**

Kaggle Silver Medal (Top 2.5%) *GitHub*

- **Model Development:** Built a **multi-resolution YOLOv8 pipeline** (384/512/768px) on 10K+ MRI scans in PyTorch with **MLflow experiment tracking**, improving severe-case sensitivity by **18%**
- **Ensemble Design:** Designed a **two-branch ensemble** (base classifier plus severe-case specialist) with cross-scale max pooling and probability normalization, cutting **weighted log-loss by 13%**; benchmarked **9 model variants** to evaluate scale-resolution tradeoffs

Publications

- T. Tang, A. Li, X. Bai, **X. Tan**, Q. Ji, M. Ulis, L. Si, L. Bao. "Bridging Data Gaps in Oncology: Integrating Large Language Models and **Recommendation Systems** for Evidence Synthesis." *Proceedings of Machine Learning Research (PMLR)*, Vol. 335, STAI-X 2026 (Statistics & Trustworthy AI, Harvard University). Poster presentation.